

MATH120-24S2 Discrete Mathematics

Tutorial 2

References

- Topics 1 and 2: Logic, Arguments and Proofs
- Chapter 4 of Epp (4th or 5th Ed.)

Work on the following problems in groups during the tutorial in week 3.

- ⇒ 1. (a) [D] Write the converse and contrapositive of the statement

$$(p \wedge q) \rightarrow r.$$

- (b) [B] Use the method of proof by contrapositive to prove that for all integers n , if n^2 is even then n is even.
- (c) [M] Can we prove that a statement of the form $p \rightarrow q$ is true by proving that its converse is false? Why/why not?

Solution:

- (a) The converse is $r \rightarrow (p \wedge q)$, while the contrapositive is $\neg r \rightarrow (\neg p \vee \neg q)$.
- (b) *Proof:* The contrapositive of the statement is: If n is odd, then n^2 is odd. Suppose n is odd, then by definition there exists an integer k such that $n = 2k + 1$. Then $n^2 = (2k + 1)^2 = 2(2k^2 + 2k) + 1$ and hence n^2 is odd, as required.
- (c) Yes. The converse is $q \rightarrow p$. Although $\neg(q \rightarrow p)$ is not logically equivalent to $p \rightarrow q$, $\neg(q \rightarrow p) \rightarrow (p \rightarrow q)$ is a valid argument. Build a truth table to see this more clearly.
2. [M] For primitive statements p, q, r and s , simplify the compound statement

$$(((p \wedge q) \wedge r) \vee ((p \wedge q) \wedge \neg r)) \vee \neg q \rightarrow s,$$

and give reasons for each step. *Note that you are not required to use truth tables, but you can do so if you wish to check your answer.*

Solution:

To simplify the compound statement

$$\left(\left((p \wedge q) \wedge r \right) \vee \left((p \wedge q) \wedge \neg r \right) \right) \vee \neg q \rightarrow s, \quad (1)$$

we begin by simplifying the statement $((p \wedge q) \wedge r) \vee ((p \wedge q) \wedge \neg r)$. This gives us

$$\begin{aligned} ((p \wedge q) \wedge r) \vee ((p \wedge q) \wedge \neg r) &\Leftrightarrow (p \wedge q) \wedge (r \vee \neg r) && \text{Distributive Laws} \\ &\Leftrightarrow (p \wedge q) \wedge \mathbb{T} && (r \vee \neg r) \Leftrightarrow \mathbb{T} \\ &\Leftrightarrow (p \wedge q). && \text{Tautology} \end{aligned}$$

Substituting this back into (1) (Rule 2) gives $((p \wedge q) \vee \neg q) \rightarrow s$. We can further simplify this as follows:

$$\begin{aligned} ((p \wedge q) \vee \neg q) &\Leftrightarrow (p \vee \neg q) \wedge (q \vee \neg q) && \text{Distributive Laws} \\ &\Leftrightarrow (p \vee \neg q) \wedge \mathbb{T} && (q \vee \neg q) \Leftrightarrow \mathbb{T} \\ &\Leftrightarrow (p \vee \neg q). && \text{Tautology} \end{aligned}$$

Therefore, statement (1) simplifies to $(p \vee \neg q) \rightarrow s$ or $(q \rightarrow p) \rightarrow s$ (since $(p \vee \neg q) \Leftrightarrow (q \rightarrow p)$).

3. [B] Determine whether the statement

$$\left((\neg p \wedge q) \wedge (q \wedge r) \right) \wedge \neg q$$

is a tautology or a contradiction by

- (a) using a truth table,
- (b) simplifying the statement using the logical equivalences in Theorem 2.8 and Rules 1&2.

Solution:

- (a) The truth table for

$$\left((\neg p \wedge q) \wedge (q \wedge r) \right) \wedge \neg q.$$

is given below.

p	q	r	$\neg p$	$\neg q$	$\neg p \wedge q$	$q \wedge r$	$((\neg p \wedge q) \wedge (q \wedge r))$	$((\neg p \wedge q) \wedge (q \wedge r)) \wedge \neg q$
0	0	0	1	1	0	0	0	0
0	0	1	1	1	0	0	0	0
0	1	0	1	0	1	0	0	0
0	1	1	1	0	1	1	1	0
1	0	0	0	1	0	0	0	0
1	0	1	0	1	0	0	0	0
1	1	0	0	0	0	0	0	0
1	1	1	0	0	0	1	0	0

Since all possible truth values of $((\neg p \wedge q) \wedge (q \wedge r)) \wedge \neg q$ are false, the statement is a contradiction.

(b) Simplifying the statement, we have

$$\begin{aligned}
 ((\neg p \wedge q) \wedge (q \wedge r)) \wedge \neg q &\Leftrightarrow (\neg p \wedge (q \wedge q) \wedge r) \wedge \neg q && \text{Associative Laws} \\
 &\Leftrightarrow (\neg p \wedge q \wedge r) \wedge \neg q && \text{Idempotent Laws} \\
 &\Leftrightarrow (\neg p \wedge r \wedge q) \wedge \neg q && \text{Commutative Laws} \\
 &\Leftrightarrow \neg p \wedge r \wedge (q \wedge \neg q) && \text{Associative Laws} \\
 &\Leftrightarrow \neg p \wedge r \wedge \mathbb{F} && q \wedge \neg q \Leftrightarrow \mathbb{F}, \text{ Rule 2} \\
 &\Leftrightarrow \neg p \wedge (r \wedge \mathbb{F}) && \text{Associative Laws} \\
 &\Leftrightarrow \neg p \wedge \mathbb{F} && \text{Contradiction} \\
 &\Leftrightarrow \mathbb{F}. && \text{Contradiction}
 \end{aligned}$$

$\Rightarrow$ 4. [M] Consider the following valid argument:

$$(((p \wedge q) \rightarrow r) \wedge \neg q \wedge (p \rightarrow \neg r)) \rightarrow (\neg p \vee \neg q).$$

- (a) Establish its validity by means of a truth table. Determine which rows of the table are crucial for assessing the validity of the argument and which rows can be ignored.
- (b) Is the statement a logical implication? In other words, does

$$(((p \wedge q) \rightarrow r) \wedge \neg q \wedge (p \rightarrow \neg r)) \Rightarrow (\neg p \vee \neg q)?$$

Why/why not?

Solution:

(a)

p	q	r	$p \wedge q$	$(p \wedge q) \rightarrow r$	$\neg q$	$p \rightarrow \neg r$	$((p \wedge q) \rightarrow r) \wedge \neg q \wedge (p \rightarrow \neg r)$	$\neg p \vee \neg q$
0	0	0	0	1	1	1	1	1
0	0	1	0	1	1	1	1	1
0	1	0	0	1	0	1	0	1
0	1	1	0	1	0	1	0	1
1	0	0	0	1	1	1	1	1
1	0	1	0	1	1	0	0	1
1	1	0	1	0	0	1	0	0
1	1	1	1	1	0	0	0	0

It's worth noting that this question can be tackled in different ways. One way is as follows: The validity of the argument follows from the results in rows 1, 2, and 5 of the table, since these are the rows in which $((p \wedge q) \rightarrow r) \wedge \neg q \wedge (p \rightarrow \neg r)$ is true. Since $(\neg p \vee \neg q)$ is also true in each of these three rows, the argument is valid. The results in the other five rows may be ignored, since they are the rows in which $((p \wedge q) \rightarrow r) \wedge \neg q \wedge (p \rightarrow \neg r)$ is false, and hence have no impact on the validity of the argument.

To confirm this, you can add the column

$$(((p \wedge q) \rightarrow r) \wedge \neg q \wedge (p \rightarrow \neg r)) \rightarrow (\neg p \vee \neg q)$$

to the truth table and check that this statement is a tautology.

You could also argue that in order to check the validity of the argument, the only rows you need to check are rows 7 and 8. These are the only rows for which the conclusion is false; therefore, in order for the statement to be valid, we simply need to have that for those rows, the premise $((p \wedge q) \rightarrow r) \wedge \neg q \wedge (p \rightarrow \neg r)$ is false as well. This is the case, so the argument is valid.

(b) From the columns in **bold** in the truth table above, we see that whenever the truth value of $((p \wedge q) \rightarrow r) \wedge \neg q \wedge (p \rightarrow \neg r)$ is 1, the truth value of $(\neg p \vee \neg q)$ is also 1. Consequently (and by definition),

$$(((p \wedge q) \rightarrow r) \wedge \neg q \wedge (p \rightarrow \neg r)) \Rightarrow (\neg p \vee \neg q).$$

5. [B] Give the reasons at each step in the following simplification of the compound statement.

$$\begin{aligned}
 \neg p \wedge (q \rightarrow p) &\Leftrightarrow \neg p \wedge (\neg q \vee p) \\
 &\Leftrightarrow (\neg p \wedge \neg q) \vee (\neg p \wedge p) \\
 &\Leftrightarrow (\neg p \wedge \neg q) \vee \mathbb{F} \\
 &\Leftrightarrow (\neg p \wedge \neg q) \\
 &\Leftrightarrow \neg(p \vee q)
 \end{aligned}$$

Solution:

The reasons for each step in the simplification of the compound statement are given below.

$\neg p \wedge (q \rightarrow p) \Leftrightarrow \neg p \wedge (\neg q \vee p)$	Implication
$\Leftrightarrow (\neg p \wedge \neg q) \vee (\neg p \wedge p)$	Distributive Law
$\Leftrightarrow (\neg p \wedge \neg q) \vee \mathbb{F}$	Contradiction
$\Leftrightarrow (\neg p \wedge \neg q)$	Contradiction
$\Leftrightarrow \neg(p \vee q)$	De Morgan's Law

6. [M] Provide an indirect proof of the statement below by stating and proving the contrapositive of the statement.

For all integers k and l , if $k + l$ is even, then k and l are both even or both odd.

Solution:

The contrapositive of the statement is: *For all integers k and l , if k and l are not both even or not both odd, then $k + l$ is odd.* Or, equivalently: *For all integers k and l , if one of k or l is even and the other is odd, then $k + l$ is odd.* *Proof:* Assume (without loss of generality) that k is even and l is odd. Then we may write $k = 2c$ and $l = 2d + 1$, where c, d are integers. Then the sum

$$k + l = 2c + 2d + 1 = 2(c + d) + 1.$$

Since $c + d$ is an integer, it follows that $k + l$ is odd.

7. [B] Prove that for every integer n , n^2 is even if and only if n is even.

Solution:

Here we must prove two results – namely, if n^2 is even, then n is even, and if n is even, then n^2 is even. *Proof:* ($\rightarrow$) Using the method of contraposition, suppose that n is not even (that is, n is odd). Then $n = 2a + 1$, for some integer a , and

$$n^2 = (2a + 1)^2 = 4a^2 + 4a + 1 = 2(2a^2 + 2a) + 1,$$

where $2a^2 + 2a$ is an integer. Hence n^2 is odd (or, not even).

($\leftarrow$) Assume that n is even. Then we may write $n = 2c$ for some integer c . So

$$n^2 = (2c)^2 = 2^2 c^2 = 2(2c^2).$$

Since $2c^2$ is an integer, it follows that n^2 is even.

8. [B] For each of the following, write a negation of the given statement, and disprove it by finding of a counterexample.

- (a) For all real numbers a and b , if $a < b$ then $a^2 < b^2$.
- (b) For all integers m and n , if $2m + n$ is odd then m and n are both odd.

Solution:

- (a) The negation of *For all real numbers a and b , if $a < b$ then $a^2 < b^2$* is *There exist some real numbers a and b such that $a < b$ and $a^2 \geq b^2$.*

Consider $a = -1$ and $b = 1$. We have $a < b$ and $a^2 = b^2$, so the statement is disproved.

- (b) The negation of *For all integers m and n , if $2m + n$ is odd then m and n are both odd* is *There exist some integers m and n such that $2m + n$ is odd but at least one of m and n is even.*

Consider $m = 4$ and $n = 3$. We have $2m + n = 11$ which is odd, but m is even, so the statement is disproved.

- $\Rightarrow$ 9. [M] Write a negation of the following statements.

- (a) Every MATH120 student lives in Christchurch.
- (b) There is a MATH120 student who likes chocolate.

Solution:

- (a) There is a MATH120 student who does not live in Christchurch.
- (b) Every MATH120 student does not like chocolate.

10. [M] Determine whether $(a + b)^2 = a^2 + b^2$ is true for all integers, true for no integers, or true for some and false for other integers. Justify your answer.

Solution:

$(a + b)^2 = a^2 + b^2$ is true for some pairs of integers.

$$\begin{aligned}(a + b)^2 &= a^2 + b^2 \\ a^2 + 2ab + b^2 &= a^2 + b^2 \\ 2ab &= 0\end{aligned}$$

This is true when at least one of a and b is 0, but false otherwise.

11. [M] Prove the following by exhaustion.
Every positive even integer less than 26 can be expressed as a sum of three or fewer perfect squares.
(For instance, $10 = 1^2 + 3^2$ and $6 = 1^2 + 1^2 + 2^2$.)

Solution:

A proof by exhaustion takes the form of a table of cases:

$$\begin{array}{llll} 2 = 1^2 + 1^2 & 4 = 2^2 & 6 = 1^2 + 1^2 + 2^2 & 8 = 2^2 + 2^2 \\ 10 = 1^2 + 3^2 & 12 = 2^2 + 2^2 + 2^2 & 14 = 1^2 + 2^2 + 3^2 & 16 = 4^2 \\ 18 = 3^2 + 3^2 & 20 = 4^2 + 2^2 & 22 = 2^2 + 3^2 + 3^2 & 24 = 2^2 + 2^2 + 4^2 \end{array}$$